Q6 – Nested Data: A Railway Timetable

Representation

Use a nested dictionary keyed first by station, then destination, with a list of train records. Each record can be a dictionary containing train number, departure time and platform. This represents one-to-many relationships and supports direct lookup by station and destination.

Operations

Iterate through .items() at each level to produce a timetable. Validate station and destination keys before lookup; use get() when a missing route should return a default. Keep times in a consistent HH:MM 24-hour format.

Complexity

Dictionary lookup is O(1) average per level. Listing all trains is O(T), where T is the number of train records. Nested dictionaries are flexible but require consistent key naming and validation.

Python Program

```
timetable = {
    "Chennai Central": {
        "Bengaluru": [{"train": "12609", "departure": "06:10", "platform": 4}],
        "Coimbatore": [{"train": "12675", "departure": "07:15", "platform": 2}]
    },
    "Egmore": {
        "Madurai": [{"train": "12637", "departure": "20:10", "platform": 5}]
    }
}
for station, destinations in timetable.items():
    for destination, trains in destinations.items():
        for train in trains:
            print(station, destination, train["train"], train["departure"], train["platform"])
```